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337614 (37)

BE (6th Semester)

Examination, Nov.-Dec., 2014

Branch : Mechanical

MACHINE TOOL TECHNOLOGY

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

ote : All questions are compulsory. Part (a) of every question is compulsory and carries 2 marks each. Attempt any 2 parts from (b), (c) and (d) of each question and carries 7 marks each. Draw neat sketches.

. 1. (a) Name the important cutting tool materials used. 2

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(b) Discuss the significance of the different elements of geometry of a single point cutting tool. 7

(c) What is a chip breaker ? Describe its different types. 7

(d) Explain the ASA type of tool angle specification system with a neat sketch. 7

Q. 2. (a) What is machinability ? 2

(b) Define tool life. Explain briefly the relationship between tool life and the variable cutting processes as per Taylor. 7

(c) Explain the tool - work piece thermocouple method with a neat sketch. 7

(d) What are the different types of cutting fluids used ? Explain briefly. 7

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Q. 3. (a) What are the various bed sections used in a lathe ? 2

(b) Explain briefly with a neat sketch the analysis of force under tail stock. 7

(c) What are the uses of reinforcing stiffner in bed. Explain. 7

(d) What is the importance of guideways in machine tool design. Explain. 7

Q. 4. (a) Differentiate between stopped drive and stepless drive with the help of a neat sketch. 2

(b) Explain the procedure for selecting speed steps in a stepped drive. 7

(c) What do you mean by productivity loss or speed loss. Explain briefly. 7

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(d) A six speed gear box is required to be designed for transmitting 10 HP with speed ranging from 400 rpm with $\phi = 1.25$. Select an optimum ray diagram and calculate the gear ratios. 7

Q. 5. (a) Name any two types of mechanial gear drives used in feed gear box. 2

(b) What are the different types of geometrical machine tool testing conducted. Explain briefly with a neat sketch any one of them. 7

(c) How is lubrication carried out for maintanence purpose of a machine tool. Explain. 7

(d) Explain Meander drive of a feed gear box with a neat sketch. 7

Note :

Q. 1.

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